

CE 101: Engineering Graphics

Course Code	CE 101
Course Name	Engineering Graphics
L-T-P-C	2-0-4-4
Prerequisites	NIL
Approval	Roorkee Course

Equivalent Course: IC 140

Course Contents

- Types of Projection, Reference Planes and Quadrants. Projection of point keeping it in different quadrants; Auxiliary planes, projection of points on auxiliary planes; Projection of lines; Oblique planes – determination of VTH when inclination are given and vice versa, conversion; Projection of plane figures; Plane figure in OP and one edge inclined to HP or VP; Types of solids and their projections in their initial positions; Section of solid and development;
- **General:** Sheet Layout, Line Symbols, Line Groups, Preferred Scales, Theory of Orthographic Projection, Technical Sketching;
- **Shape Description (External):** Multiplaner Representation Systems of Projection, Sketching of Orthographic Views from Pictorial Views, Conventional Practices, Precedence of Views, Precedence of Lines;
- **Uniplaner Representation:** Sketching of Pictorial Views(Isometric and Oblique) from Multiplaner Orthographic Views;
- **Shape Description (Internal):** Sectioning as an Aid to understand internal features, Principles of Sectioning, Types of Sections, Section Lines, Cutting Plane Lines and Conventional Practices;
- **Size Description:** Dimensioning, Tools of Dimensioning, Size and position Dimensions, Unidirectional and Aligned Systems, Principle and Practices, of Dimensioning, Tolerance Dimensioning;
- **Conventional Representation:** Representation and Identification of Common Machine Elements and Features
- **Practicals:** Projection of Points; Projection of lines; Oblique planes; Projection of Plane Figures; Projection of Solids; Section and Development; Sketching of Orthographic; Views from Pictorial Views; Sketching of Pictorial Views (isometric and Oblique) from Multiplaner Orthographic Views; Missing Lines Exercise; Missing View Exercise; Sectioning Exercise; Dimensioning Exercise; Identification Exercise.

References

1. Giesecke, Mitchell, Spencer, Hill, Dygdon and Novak, **Technical Drawing**, Macmillan Publishing Company
2. French T. E., Vierck C. J. and Foster R. J., **Engineering Drawing and Graphics Technology**, McGraw-Hill Inc
3. Luzadder W. J., Warren J. and Duff J. M., **Fundamentals of Engineering Drawing**, Prentice Hall international Editions
4. Sp 46:1988 **Engineering Drawing Practice for Schools and Colleges**, Bureau of Indian Standards
5. Chandra A. M. and Chandra S., **Engineering Graphics**, Narosa Publishing House.